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(54) **METHOD FOR PROCESSING A LAYER ELEMENT**

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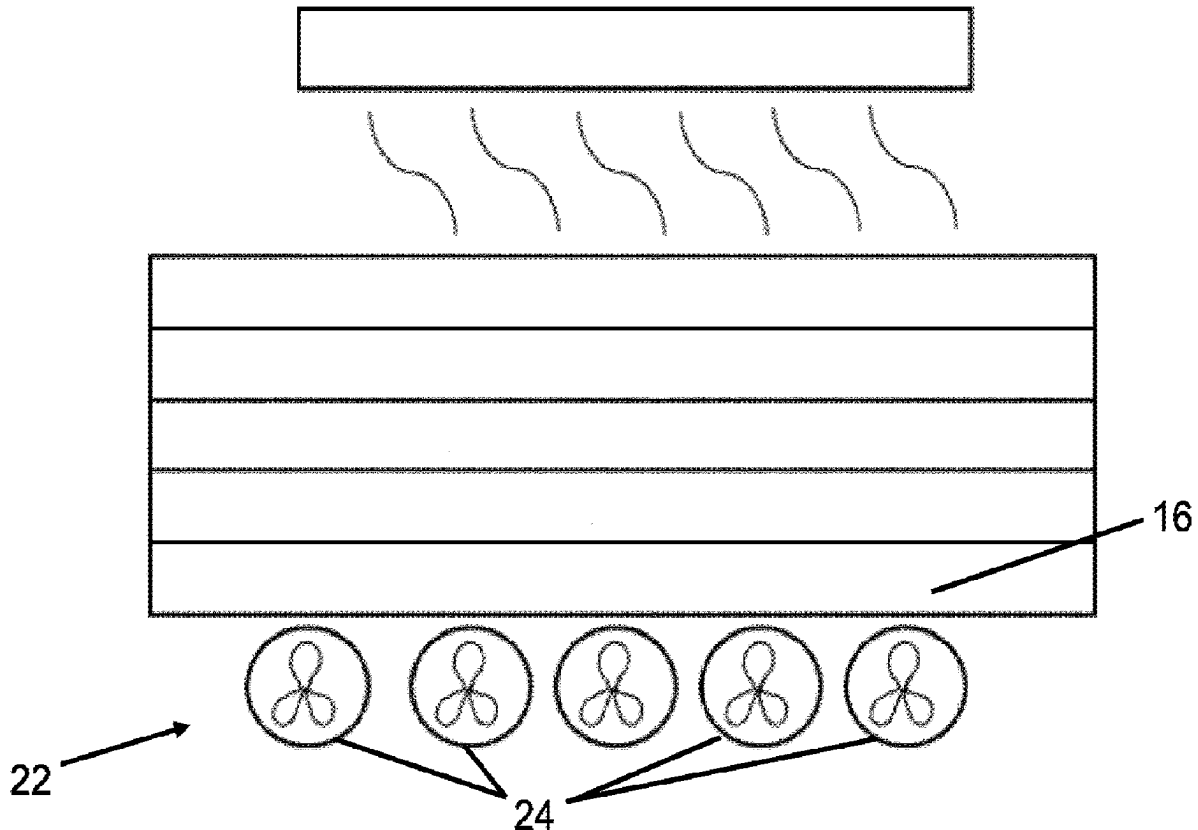
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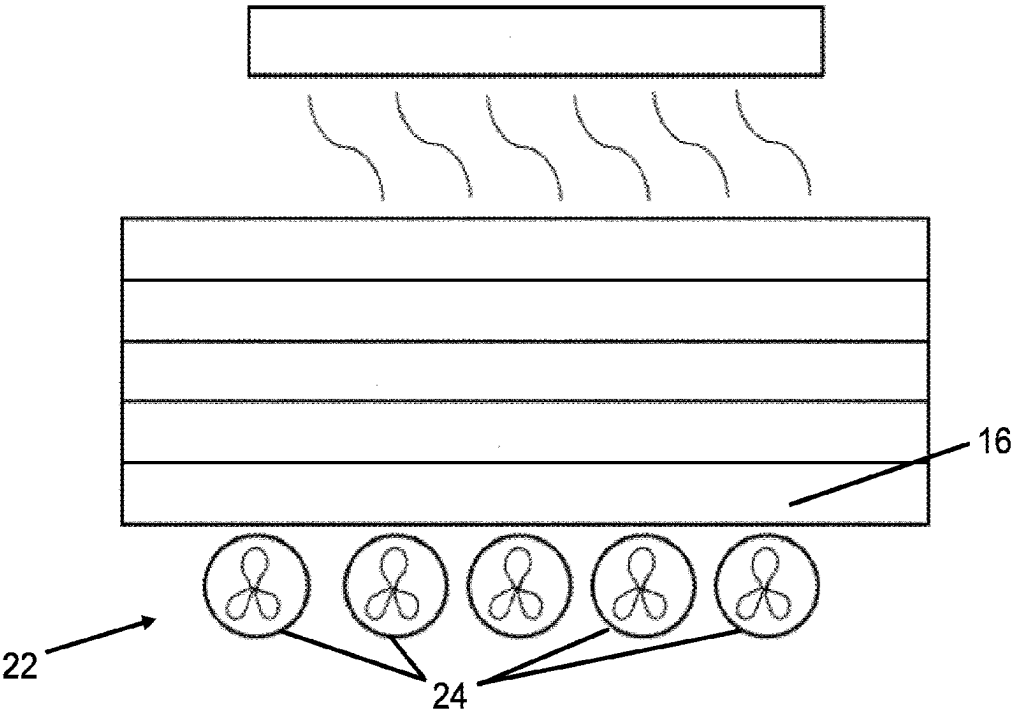
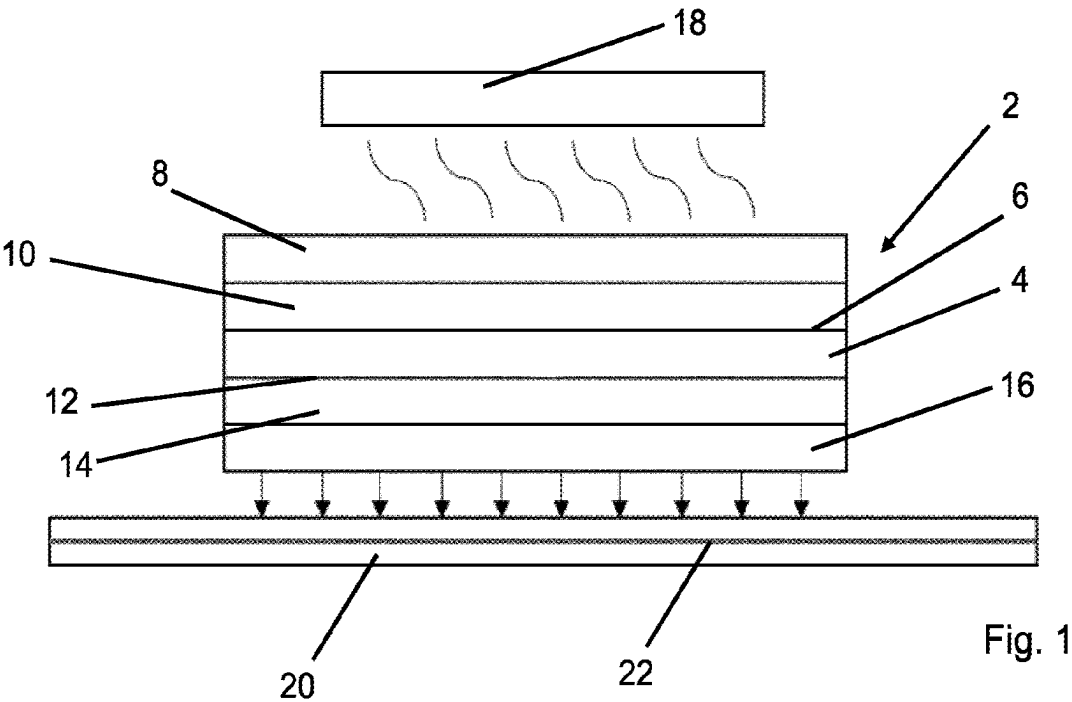
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ABSTRACT

The invention relates to a method for processing a layer element, wherein the layer element has at least one base body with a lower side and an upper side opposite the lower side, a cover layer that is at least partially transparent to visible light, and at least one adhesive layer arranged between the cover layer and the upper side, wherein, during the method, the adhesive layer is heated to an adhesive layer temperature and the lower side of the base body is cooled to a lower side temperature.

Specification includes a Sequence Listing.





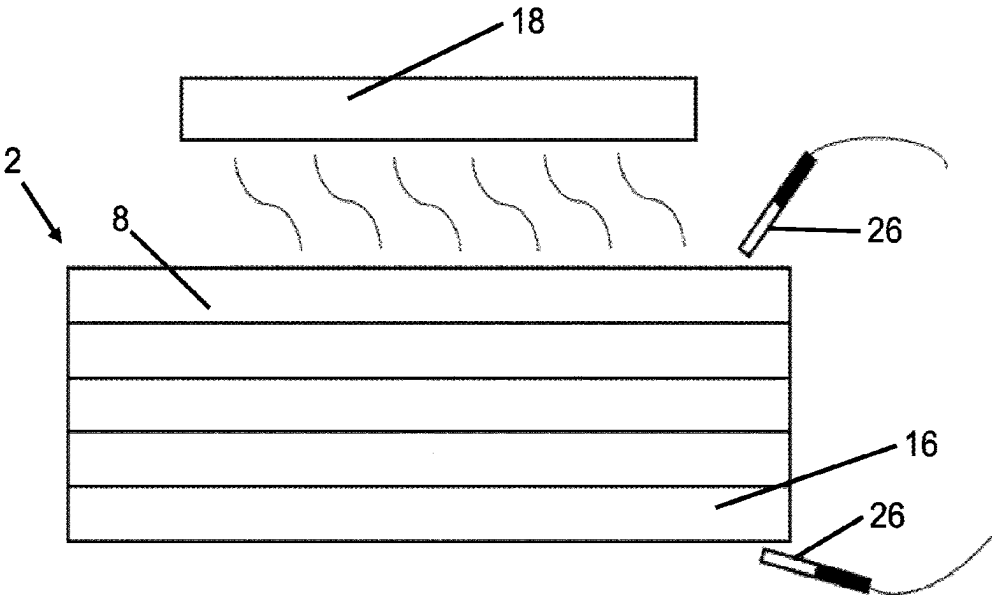


Fig. 3

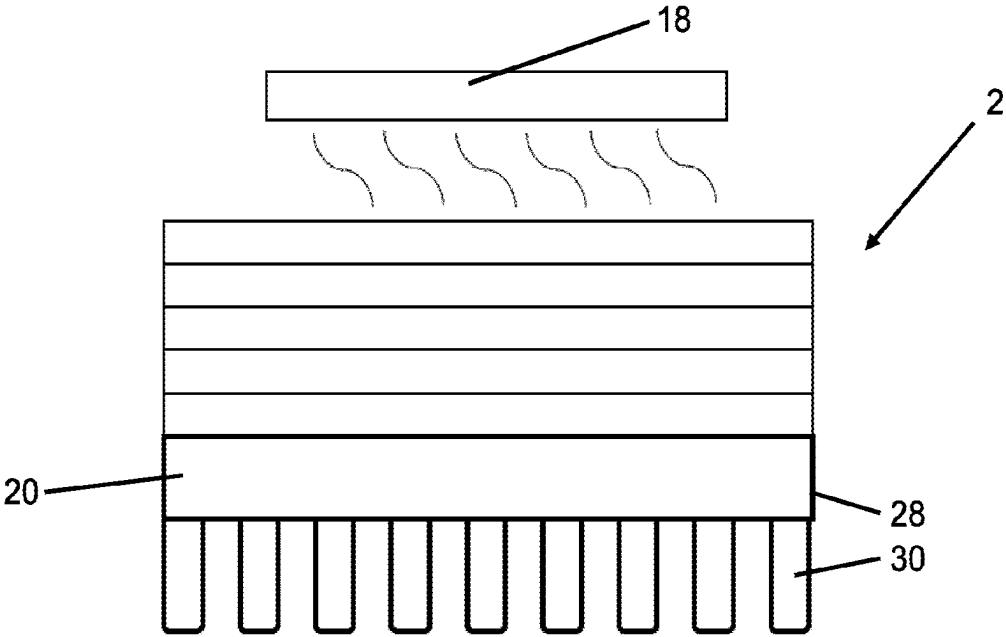
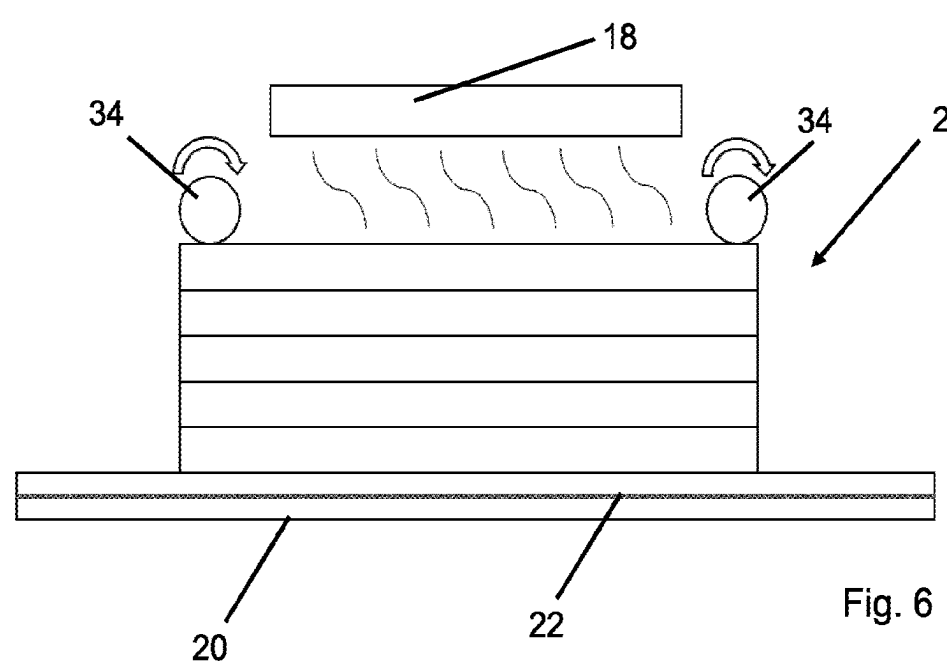
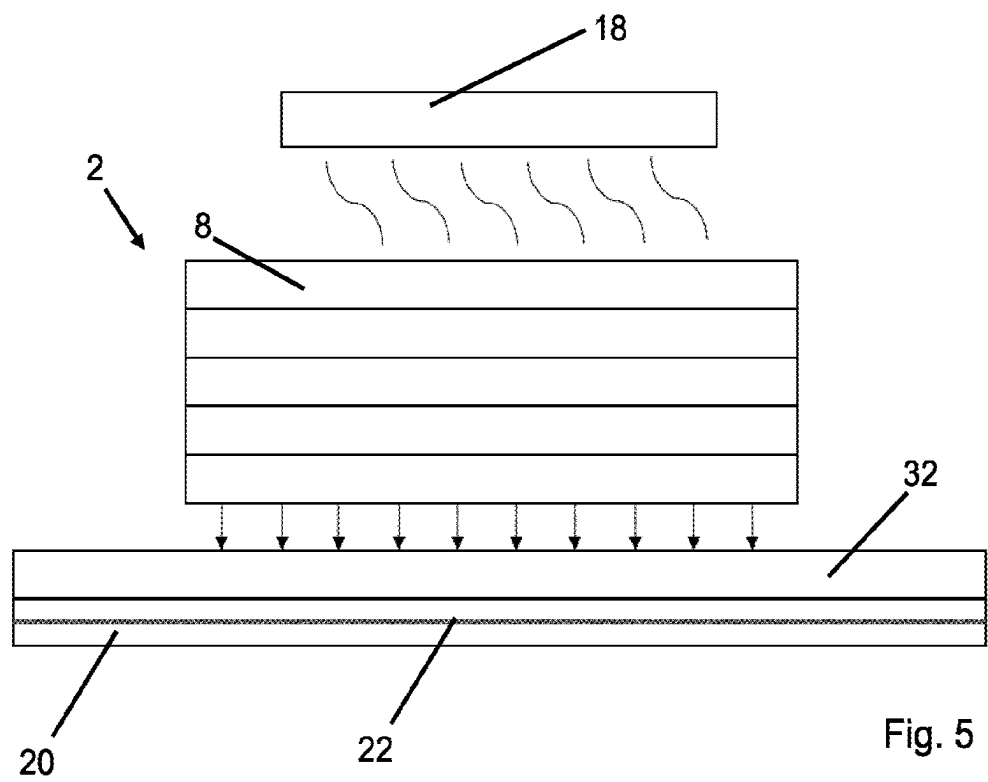


Fig. 4



METHOD FOR PROCESSING A LAYER ELEMENT

[0001] The invention relates to a method for processing a layer element, wherein the layer element has at least one base body with a lower side and an upper side opposite the lower side, a cover layer that is at least partially transparent to visible light, and at least one adhesive layer arranged between the cover layer and the upper side. The invention also relates to a device for conducting such a method.

[0002] In the context of the present invention, a layer element is, for example, a solar module comprising said layers. Other components and elements comprising said layers are likewise to be understood under the term layer element. Solar modules of the type mentioned above are being used in increasing numbers to generate electricity from sunlight. For example, said solar modules may be placed in small numbers on roofs of private buildings, such as residential buildings, in order to at least partially meet the electricity demand of the house on which the solar modules are mounted. This use of this kind of decentralized energy supply is increasing, for example to charge electric vehicles decentrally. However, solar modules of the type mentioned above are also used in large installations on otherwise agricultural land to generate commercial electricity and feed it into the public grid.

[0003] The disadvantage is that the solar modules only have a limited service life. Expensive raw materials are used during production of the solar modules. For example, the conductor tracks are manufactured from a large number of modules made of silver. For this reason alone, the prior art describes methods that can recycle said raw materials and make them available for reprocessing. For solar modules which contain metal components arranged on a glass substrate, JP 2014/054593 A, for example, proposes a method in which the components are mechanically removed. For solar modules with a silicon substrate, the previously non-published DE 10 2021 129 301 discloses a method in which the adhesive layer arranged between the cover layer and the conductor tracks is heated so as to enable the cover layer to be removed from the rest of the solar module.

[0004] The disadvantage is that this method also cannot be used for all solar modules with a silicon base body. The invention is therefore based on the task of improving the method in such a way that it can be applied for a wider range of different solar modules.

[0005] The invention solves the addressed task by way of a method according to the preamble of claim 1, characterized in that, during the method, the adhesive layer is heated to an adhesive layer temperature and the lower side of the base body is cooled to a lower side temperature. Heating the adhesive layer makes it possible to remove the cover layer from the rest of the layer element. The cover layer is preferably a glass layer which, particularly preferably, is designed to be completely transparent for visible light. By heating the adhesive layer, the adhesive force of the adhesive layer decreases and the cover layer can be removed. The heat introduced into the adhesive layer spreads inside the layer element and gradually heats other layers and elements of the layer element. A plastic layer, for example of polyvinyl fluoride (PVF), is often arranged on the lower side of the base body. The material of this plastic layer is usually less heat-resistant than the base body, the cover layer and especially the adhesive layer. The heat introduced to heat the adhesive layer also heats the plastic layer and can cause it to

disintegrate, which can release gases that are harmful and pose a health hazard. According to the invention, this is prevented by cooling the lower side of the base body to a lower side temperature.

[0006] Within the context of the present invention, processing a layer element comprises, for example, at least partially dismantling the layer element. In one preferred embodiment, this is achieved by either completely or partially removing the cover layer from the base body. This is preferably done without damaging or destroying the cover layer in the process. However, this is not necessarily the case. Even if the cover layer is damaged or destroyed during complete or partial removal from the base body and/or prior to complete or partial removal from the base body, it still constitutes "processing" within the context of the present invention.

[0007] This does not mean that the lower side is cooled during the method to a temperature below room temperature, for example. Prior to starting the method, the lower side of the base body of the layer element is at a temperature that is significantly lower than the lower side temperature that is aimed for during the method. All it means is that heat is dissipated from the lower side of the base body in order to achieve and preferably maintain the lower side temperature. The lower side temperature is lower than the adhesive layer temperature.

[0008] In order to cool the lower side of the base body to a lower side temperature, a cooling device is provided to dissipate heat from the lower side of the base body.

[0009] One difficulty that is eliminated by the method according to the invention is that the conductor tracks of the layer elements, for example of solar modules, which are produced from the valuable raw material, are not accessible from the outside. They are located on the upper side of the base body, but are covered by the cover layer, which is fixed by the at least one adhesive layer. Heating the adhesive layer reduces its adhesive effect so that the cover layer and base body can be removed from each other afterwards. The conductor tracks arranged on the upper side of the base body are then accessible. They are now removed from the base body by mechanical removal and the removed material is collected.

[0010] The layer element preferably refers to a solar module, wherein the layer element preferably comprises metallic conductor tracks which, particularly preferably, are arranged on the upper side of the base body and are covered by the cover layer.

[0011] Preferably, the adhesive layer temperature is at least 180° C., preferably at least 200° C., especially preferably at least 230° C., and at most 400° C., preferably at most 350° C., especially preferably at most 280° C. This is particularly advantageous if the adhesive layer is a layer made of or with ethylene-vinyl acetate (EVA). The adhesive layer temperature is preferably selected in such a way that pyrolysis of the adhesive layer does not occur, which can release partially poisonous and environmentally harmful gases. This is prevented by a temperature below 400° C. At a temperature of more than 180° C., preferably more than 200° C., acetic acid in an EVA adhesive layer is released, which creates a lubricating film between the layers connected by the adhesive layer. This causes forces of adhesion, which are applied by the adhesive layer, to decrease and the

two connected elements can be separated from each other. Preferably, therefore, at least one EVA film is also used as an adhesive layer.

[0012] The lower side temperature is preferably at most 150° C., preferably at most 110° C., especially preferably at most 100° C. The lower this temperature, the smaller the effects of the heat on layers arranged on the lower side of the base body.

[0013] The base body preferably has a structure composed of multiple layers arranged next to each other. The lower side of the base body constitutes the lower side of the cover element that is processed using the method described here. For example, the base body has a substrate, which is preferably produced from silicon, particularly preferably made of silicon, and a charge-doped zone and a hole-doped zone. The metallic conductor tracks are arranged on the upper side of this substrate, which forms the upper side of the base body. The lower side of the substrate does not necessarily have to form the lower side of the layer element. Further layers, such as the previously mentioned plastic layer, can be arranged on the lower side of the substrate. The lower side of the base body preferably forms the lower side of the layer element.

[0014] In one preferred embodiment, the adhesive layer is heated by means of infrared radiation and/or magnetic induction and/or microwave radiation. In particular, it is advantageous when using infrared radiation or microwave radiation to apply the radiation through the cover layer, which is at least partially transparent, but preferably completely transparent, to the respective type of radiation. Infrared radiation used to heat the adhesive layer preferably has a wavelength between 1000 and 4000 nm, preferably from 1000 to 2000 nm.

[0015] If microwave radiation is used, it preferably has a frequency of between 2.4 GHz and 2.5 GHz, preferably 2.45 GHz. Alternatively, the microwave radiation has a frequency of 5.8 GHz.

[0016] During heating by means of magnetic induction, alternating magnetic fields are used which preferably have a frequency of between 1 kHz and 500 kHz. A suitable frequency range is referred to as a low frequency range and encompasses frequencies between 1 kHz and 7 kHz. A different frequency range is referred to as a medium frequency range and contains frequencies between 8 kHz and 40 KHz. The high frequency range contains frequencies between 60 kHz and 500 kHz. The frequency used is preferably between 25 kHz and 300 kHz, especially preferably between 25 kHz and 100 KHz.

[0017] The invention also solves the addressed task by way of a device for carrying out one of the methods described here, characterized in that the device comprises a heating device for heating the adhesive layer of a layer element to the adhesive layer temperature and a cooling device for cooling the lower side of the base body of the layer element to the lower side temperature.

[0018] Preferably, the device has a holding device with which the layer element processed using the device is held. The device preferably has a workbench for supporting the layer element, the cooling device preferably being arranged within the workbench. In a particularly preferred embodiment, the workbench has a contact surface on which the layer element rests while the method is conducted. This contact surface can preferably be cooled by the cooling device. The holding device with which the layer element is

held is, for example, a traction device for exerting a tensile force. It may comprise, for example, a negative pressure element, such as a suction element. The layer element is positioned, for example, on openings within the contact surface, which are sealed by the layer element. By applying a negative pressure or a suction force to the sealed openings, a suction force and therefore a tensile force is exerted on the layer element that holds it on the contact surface.

[0019] Alternatively or additionally, the holding device has a pressure device for exerting a compressive force. This compressive force causes the layer element to be pressed onto the contact surface. The pressure device preferably comprises at least one, preferably multiple, hold-down clamps. They can preferably be moved relative to the contact surface. The hold-down clamps are preferably removed from the contact surface in order to arrange a layer element to be processed on the contact surface. The hold-down clamps are then lowered until they come into contact with the layer element and are able to apply a compressive force to the layer element.

[0020] Preferably, the cooling device has at least one fluid channel, through which a coolant can be directed. The coolant is, for example, a cooling liquid such as water. The cooling device preferably has a pump, by means of which the coolant can be moved through the fluid channel. If the fluid channel is situated within the workbench, it is advantageous for said workbench to be made of a material with high thermal conductivity, for example a metal, such as aluminium or steel.

[0021] The at least one cooling channel is preferably located as close to the actual contact surface of the workbench as possible, for example at most 15 cm, preferably at most 10 cm, especially preferably at most 5 cm, away from the contact surface. The material of the workbench, at least the material of the workbench between the at least one cooling channel and the contact surface, preferably has a high thermal conductivity. The material is preferably a metal, such as aluminium or steel.

[0022] Alternatively or additionally, the cooling device has at least one fan through which the air can be directed onto the lower side of the layer element. The fan may be designed, for example, in the form of one or multiple ventilators, which are arranged and aligned such that they move air towards the lower side of the layer element.

[0023] Alternatively or additionally, the cooling device has at least one device for discharging compressed air. This device is configured such that the compressed air, which comes from a compressed air source that is preferably part of the cooling device, but is at least part of the device, is directed onto the lower side of the layer element. In the methods described here, the lower side of the layer element is preferably cooled via the fan and/or the device for discharging compressed air.

[0024] The device preferably has a first temperature sensor for detecting the adhesive layer temperature and/or a second temperature sensor for detecting the lower side temperature. Preferably, the first temperature sensor and/or the second temperature sensor have/has a pyrometer by means of which the temperatures that occur can be detected, preferably in a contactless manner.

[0025] The device preferably has an electronic control unit, especially preferably an electronic data processing device, which is configured to control the heating device and/or the cooling device depending on the detected adhe-

sive layer temperature and/or the detected lower side temperature. To this end, the respective detected temperature is compared with a target temperature stored in an electronic memory. The target temperature may also be a temperature range, within which the detected temperature should lie. The electrical control unit is configured in such a way that it increases the power of the heating device and/or increases the duration of heating if the detected adhesive layer temperature is considered too low after comparison with a stored target temperature. Alternatively or additionally, the electrical control unit is preferably configured in such a way that it increases the power of the cooling device and/or increases the duration of cooling if the detected lower side temperature is considered too high after comparison with a stored target temperature.

[0026] In the following, a number of embodiment examples of the invention will be explained in more detail with the aid of the accompanying drawings. They show:

[0027] FIGS. 1 and 6—schematic views of a layer element in the method.

[0028] FIG. 1 depicts a layer element 2, which is designed as a solar module. It has a base body 4, on the upper side 6 of which conductor tracks are located, the latter not being depicted in the figures. The layer element 2 also has a cover layer 8 made, for example, of glass. The cover layer 8 is arranged on the base body 4 by means of an adhesive layer 10. A reverse side 16 is arranged on the lower side 12 of the base body 4, which is opposite the upper side 6, by means of a further adhesive layer 14. During the method, the layer element 2 should be processed, for example, by recycling it. To this end, the cover layer 8 must first be removed. To achieve this, the adhesive layer 10 is heated. In the embodiment example in FIG. 1, a heating unit 18 is used for this purpose. The upper side of the layer element 2 is heated via schematically depicted thermal radiation. Given that the cover layer 8 is at least partially, but preferably completely transparent for the radiation, which may also be visible light, such as laser radiation, the adhesive layer 10 is heated by the thermal radiation and thus loses at least part of its adhesive effect.

[0029] The reverse side 16 of the layer element 2 comes into contact with a workbench 20, in which a cooling device 22 is located. The latter is schematically represented as a solid line and may, for example, comprise cooling ducts through which a cooling medium is directed.

[0030] The only difference between FIG. 1 and FIG. 2 is a modified cooling device 22. In FIG. 2, it is designed in the form of multiple fans 24, by which air is directed to the reverse side 16 of the layer element. This can be achieved, for example, by way of openings in the workbench.

[0031] FIG. 2 depicts the layer element 2 as well as the heating unit 18. To be able to determine the temperature of the cover layer 8 and the reverse side 16, two temperature

sensors 26 are provided. They can be used to measure the corresponding temperatures and transmit the measurement data to an electric control unit, not depicted, which controls the heating device and/or the cooling device.

[0032] In the embodiment shown in FIG. 4, the layer element 2 rests on the workbench 20, which is designed as a cooling plate 28. Said cooling plate 28 forms part of the cooling device 22. In the embodiment example shown, the cooling device 22 also has cooling fins 30, which are arranged on the side of the workbench 20 facing away from the layer element 2. As a result, heat that is transferred from the layer element 2 to the workbench 20 and the cooling plate 28 can be discharged. The heat is preferably transferred to the surrounding air. To this end, the cooling fins 30 are preferably cooled by at least one, but preferably multiple fans, which are not shown in FIG. 4.

[0033] FIG. 5 shows a further embodiment. The layer element 2 is positioned on the workbench 20 in which the cooling device 22 is located, as in FIG. 1. In the embodiment example shown, a vacuum plate 32 is arranged between the layer element 2 and the part of the workbench 20 containing the cooling device, said vacuum plate applying a suction force to the layer element 2 and thereby firmly holding it on the workbench 20, of which the vacuum plate 32 is preferably a part.

[0034] FIG. 5 largely corresponds to FIG. 1. Conversely to this, in FIG. 6 the workbench 20 is not equipped with a holding device, such as a suction device. As such, the layer element 2 can be moved on the upper side of the workbench 20, for example to change the point heated by the heating unit 18. Rollers 34 are used for this purpose, which are depicted on the upper side of the layer element 2. The rollers can be rotated in two directions, one of which is illustrated by the arrow shown in FIG. 6.

REFERENCE LIST

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[0036]	4 base body
[0037]	6 upper side
[0038]	8 cover layer
[0039]	10 adhesive layer
[0040]	12 lower side
[0041]	14 adhesive layer
[0042]	16 reverse side
[0043]	18 heating unit
[0044]	20 workbench
[0045]	22 cooling device
[0046]	24 fan
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[0051]	34 roller

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aattacgccc	gcgcctgcca	ctcatcgcat	tactgttgta	attcattaag	cattctgcgc	8280
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gccacgttta	aatcaaaact	gggtgaaact	acccagggat	tggtcgagac	gaaaaacata	8460
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gctcatagac	cgtatccaaa	gcacccgggg	ctgatccccg	gcgccagggt	gggtttttct	9060
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gcgggatata	acatgagctg	tcttcgggtat	cgtcgtatcc	cactaccgag	atatccgcac	9240
caacgcgcag	cccggaactg	gtaatggcgc	gcattggccc	cagcgcac	tgatcgttgg	9300
caaccagcat	cgcagtgagg	acgatgccct	cattcagcat	ttgcatgggt	tggtgaaaac	9360
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gatatttatg ccagccagcc agacgcagac gcgcccagac agaacttaat gggcccgccta 9480
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aaatattctg aaatgagctg ttgacaatta atcatcggtt cgtataatgt gtggaattgt 10380
gagcgggataa caatttcaca caggaaacag aatt 10414

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SEQ ID NO: 11      moltype = DNA length = 1443
FEATURE            Location/Qualifiers
source              1..1443
                    mol_type = genomic DNA
                    organism = Streptomyces violaceoruber

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SEQUENCE: 11
atgcgctcgc acctgctcaa cgacaccacc gcggagcagt accgcgcgttc cgtgaccgaa 60
ggagtcgagc ggggtggcgc caaactcgcc accaccgacc gcccggtcac cggcgtcacg 120
gtcgacgccc tctccccccg catcgacgcy atcgacctcg acgagccgct gcacgacacc 180
gccgcggtcc tcgacgagct ggaggacgtc taactccgcy acgcccgtcta cttccaccac 240
ccccgctacc tcgcccact caactgccc gtcgtcatac cggcgctgct cgggtgaagcg 300
gtcctgtccg ccgtcaactc ctccctggac acctggggacc agtcggccgg cggcacgctc 360
atcgagcgga agctgatcga ctggacctgc gcccgcatcg gcccgccc gccggccgac 420
ggcggtgtca cctccggcgg caccagtc aaactccagg cgtgctcct cggccgagc 480
gaggcggaag ccgaggactt cgcgcacctg cggatcttcg cctccgaggg cagccacttc 540
agcgtcagga agtcgcgcaa actgctcgcc ctcggccccc acgcccgtgt gtcgatcccg 600
gtcgaccgcy acaagcggtt gcagaccgtc gccctcgccc gcgagctgga ggcgtgcgcy 660
cgggacgccc tggcccccat ggcgctcgtc gccaccggcg gcaccaccga cttcggtcgt 720
atcgaccgcy tcgcggagat cgcggggctg tgcgagcagt acgcccgtgt gatgcacgtc 780
gacgcgccct aggcgtgcgg gctgctcgcc tccctgaagt accgggaccg catcacggcg 840
atcgagcggy ccgactcggt caccgtggac taccacaagt cctctctcca gccggtgagt 900
tcgtcgcccg tgctgggtcg ggaagcgggc acctgcccgc acgccacctc ccacgcgga 960
tacctcaacc cgcgcgcat ggtgcaggaa cgtatcccca accaggtgga caagtcctc 1020
cagaccaccc gccgcttcga cgcgctcaag ctgtggatga cgtgcgctg gatggcgcc 1080
gacggcatcg ggcctctctt cgacgaggtg tgcgacctgg ccgcccaggc ctggaaactg 1140
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cactacctga agttcacctt gctcaacccc gagacgaccc cggccgacat cgcgcgctc 1380
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tga 1443

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SEQ ID NO: 12      moltype = DNA length = 1281
FEATURE            Location/Qualifiers
source              1..1281
                    mol_type = genomic DNA
                    organism = Streptomyces violaceoruber

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SEQUENCE: 12
ttgaccgcyg ttctgaagc cagcgccccg tacgacttcg tgggcatcgg cctcgcccc 60
ttcaacctcg gctcgcgtg cctcaccgag cccgtcgccg agctgaacgg cgtcttctc 120
gagtcacaag cggacttcga gtggcacgcc gggatgttcc tggacggcgc ccacctccag 180
acccggttca tgtcgacct ggtgacgctc gccgaccga cctcgccgta ctcttctc 240
aactacctga aggagcagg cggcggtgac tcttctaca tccgggagaa cttctaccg 300
ctgcccgtcg agtacgacga ctactgccgc tgggcccgcg gcaagctgag cagcgtccgc 360
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accacgtccg gcgacacctc cgcgccccgc cgcctcgtcc tcggcaccgg caccgccgcg 480
cacatccggg acgcctgcgg gggcctggcc ggcgacttcc tccacaactc ccggtacgcy 540
cggcaccggg cggagctggt gaagaagaag tcgatcacgc tggtcggcag cggccagtcc 600
gccgcccaga tctaccagga cctgctgagc gagatcgagc tccaccgcta cgggtgtaac 660
tgggtgaccc gctccccgcy gttcttccc ctcgaataca ccaagctcac cctggagatg 720
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acggccgagc agaagggcct gttcaagggc atcgacggcg acctgatcaa cgagatcttc 840
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gcgctgacca gcgcccggta tgcggacggc acctacacgc tcgggttcg ccaggaggag 960
cagggcacgg acttcagat cgagaccgag ggcctggctc tcgccaccgg ctaccggtac 1020
accgagccgy agttctcaa gcccgctcgg gaccggctgc gctacgactc ccgcccgaac 1080
ttcgacatcg gccgcaacta cgcgctcgac gtcacgggag gcggcggtgt cctccagaac 1140
gccccgggtcc acgcgcacag cgtcaccagc cccgacctgg gcatggcgcy ctaccgcaac 1200

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agctgcacgc tccgggagct gctcggacgc gagtactacc cggtcgagca gtcgatcgcg 1260
ttccaggagt tcgcggtatg a 1281

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SEQ ID NO: 13      moltype = DNA length = 555
FEATURE           Location/Qualifiers
source            1..555
                  mol_type = genomic DNA
                  organism = Streptomyces violaceoruber

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SEQUENCE: 13
atgagccgct tggagcaccac cccccccgtc ggggcactga ccctgcgccc cgtcgacccg 60
ctgacgggac cgtactgtct gcacggctgg ctaccccacc ccaagtcgcg gttctggatg 120
atgcaggacg cccggctggt ggacgtcgag cgggcctaca tggagctggc cgcgcacgag 180
caccagcagg cccacctcgg cctgcacgac ggggtcccg ccttctctgac ggagcgctac 240
gaccccgccc acccggaact ggtcgggctg tacgagcccg agccgggcga cgtcggcatg 300
cacttctctg tcgcgcccac cgaccggccc gtgcacggct tcacccgccc cgtgatcacc 360
accgtgatga cggagctgtt cgcgcacccg gcgacccggc gggtcgtcgt cgaaccggac 420
gtcaccaca cgcgcgtgca cgcctgaac gcagcgtcgc gattcgtgcc cgagcgcgag 480
atccagaagc cggagaagaa ggccttgctg agcttctgca cccgcgagca gttcgcgaag 540
gcggtgtccg catga 555

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SEQ ID NO: 14      moltype = DNA length = 1788
FEATURE           Location/Qualifiers
source            1..1788
                  mol_type = genomic DNA
                  organism = Streptomyces violaceoruber

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SEQUENCE: 14
atgagcctcg ccgacgcgct cgcaccctcg acccccgaa cgtgggagga ggccaaccgc 60
ctcctgggtc gcaaggcgct ggcagagttc acccagagc ggctgctgac ccccgagcgg 120
gagccgggac cggggggcgg gcagacgtac gtcgtccgca gcgacgacgg ccagaccgcg 180
taccgcttca cgcgccacct cgcgcgccct gaccactggc aggtggagcg cgcctccgtc 240
accgcgccac gcgacggcgc ggaactcccg ctgcgcgcgc tggactttct catcgagctg 300
aagcagaccc tgggcctgga cgacgagatc ctcccgggtc acctggagga gatctcctcc 360
accctctccg gcaactgcta caaactgacc aagccgcgag tcagctccgc cgagctggcc 420
cggagcggag acttccaggc cgtcgagacg ggcagtgacc agggccaccc gtgcttgcgc 480
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gcgagcccgg tccggctggt gtggctggcc gcgcaccgct cgcgggcggc gttcacggcg 600
ggcgtgggca tcgagtaaga gtcttctgct cgggacgagc tgggcgcggc caccgtcgac 660
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gagctggccc acaagttcga cgcgtacgac atgttcgcgc ccgagttcgc cctgtcctgc 1680
ctcaaccggc tccagctgcg cgacaaccgg cagatggtgg acctggccga cccgtccggc 1740
gcgctccagc tcgtcggcac cctgaagaac cccctggcag gccggtag 1788

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SEQ ID NO: 15      moltype = AA length = 480
FEATURE           Location/Qualifiers
source            1..480
                  mol_type = protein
                  organism = Streptomyces violaceoruber

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SEQUENCE: 15
MRSHLLNDTT AEQYRRSVTE GVERVAAKLA TTDRPFTGVT VDALSPRIDA IDLDEPLHDT 60
AAVLDELEDV YLRDAVFPHH PRYLAHLNCP VVIPALLGEA VLSAVNSSLD TWDQSAGGTL 120
IERKLIDWTC ARIGLGPAA DGVFTSGGTQS NLQALLLARE EAKAEDFADL RIFASEASHF 180
SVRSKAKLLG LGPDVVSIIP VDRDKRMQTV ALARELERCA RDGLVPMMAV ATGGTTDFGS 240
IDPLPEIAGL CEQYGVVMHV DAAYGCGLLA SLKYRDRITG IERADSVTV D YHKSFFQPV 300
SSAVLVRDAA TLRHATYHAE YLNPVRMVQE RIPNQVDKSL QTTRRFDAK LWMTLRVMGA 360
DGIGVLFDV CDLAAEGWKL LAADPRFDV VQPSLSTLVF RHIPADVTD AEIDRANLYA 420
RKALFASGDA VVAGTKVAGR HYLKFTLLNP ETPADIAAV LDLIAGHAEQ YLGDSLDRAS 480

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SEQ ID NO: 16      moltype = AA length = 420
FEATURE           Location/Qualifiers
source            1..420
                  mol_type = protein

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                                organism = Streptomyces violaceoruber
SEQUENCE: 16
LTALPEASAP YDFVGIGLGP FNLGLACLTE PVAELNGVFL ESKPDFEWHA GMFLDGAHLQ 60
TPFMSDLVTL ADPTSPYSFL NYLKEQGRLY SFYIRENFYP LRVEYDDYCR WAARKLSSVR 120
FSTTVTEVTY DEREELYAVA TTSGDTYRAR RLVLGTGTPP HIPDACRGLA GDFLHNSRYV 180
RHRAELVKKK SITLVGSGQS AAEIYQDLLS EIDVHGYGLN WVTRSPPFFP LEYTKLTLEM 240
TSPEYVDYH ALPEDTRYRL TAEQKGLFKG IDGDLNIEIF DLLYQKRLGG PVPTRLTNS 300
ALTSARYADG TYTLGFRQEE QGTDFFEIETE GLVLATGYRY TEPEFLKFPVR DRLRYDSRGN 360
FDIGRNYAVD VTGGGVFLQN AGVHAHSVTS PDLGMGAYRN SCIVRELLGR EYYPVEQSLA 420

SEQ ID NO: 17                moltype = AA length = 184
FEATURE                      Location/Qualifiers
source                       1..184
                               mol_type = protein
                               organism = Streptomyces violaceoruber

SEQUENCE: 17
MSRLSTTTPV GALTLPVPDP LTDAVLLHGW LTHPKSAFWM MQDARLVDE RAYMELAAD 60
HQAHLGLHD GVPALFTERY DPAHRELVLG YEPEPGDVGM HFLVAPTRDP VHGFTRAVIT 120
VTMTELPADP ATRRVVPEPD VTNTAVHALN AAVGVFPERE IQKPEKKALL SFCTRQFAK 180
AVSA 184

SEQ ID NO: 18                moltype = AA length = 595
FEATURE                      Location/Qualifiers
source                       1..595
                               mol_type = protein
                               organism = Streptomyces violaceoruber

SEQUENCE: 18
MSLADAVAH LTPERWEEANR LLVRKALAEF THERLLTPER EPDDGGGQTY VVRSDDGQTA 60
YRFTATVRAL DHWQVDAASV TRHRDGAELP LAALDFFIEL KQTLGLSDEI LPVYLEEISS 120
TLSGTCYKLT KPQLSSAELA RSGDFQAVET GMTGHPFCFV ANNGRLGFGI HEYLSYAPET 180
ASPVRLVWLA AHRRAAFTA GVGIEYESFV RDELGAATVD RFHGVLRGRG LDPADYLLIP 240
VHPWQWKNL TVTFAAEVAR GHLVCLGEGD DEYLAQQSIR TFFNASHPGK HYVKTALSVL 300
NMGFMRGLSA AYMEATPAIN DWLARLIEGD PVLKETGLSI IRERAAYGYR HLEYEQATDR 360
YSPYRKMLAA LWRESPPVSI REGETLATMA SLVHQDHEGA SFAGALIERS GLTPTEWLRH 420
YLRAYYVPLL HSPFYAYDLVY MPHGENVILV LADGVVRRAY YKDIAEEIAV MDPDAVLPPE 480
VSRIADVDPD DKLLSIFTD VFDCFFRFLA ANLAEEGIVT EDAFWRTVAE VTREYQESVP 540
ELADKFERID MPAPEFALSC LNRLQLRDNR QMVDLADPSG ALQLVGT LKN PLAGR 595

SEQ ID NO: 19                moltype = DNA length = 5120
FEATURE                      Location/Qualifiers
misc_feature                 1..5120
                               note = Vector Construct
source                       1..5120
                               mol_type = other DNA
                               organism = synthetic construct

SEQUENCE: 19
acaggaaaca gaattaatta agcttcccca ccggaggacc cccccatgcg ctcgcacctg 60
ctcaacgaca ccaccgcgga gcagtagccg cgttccgtga ccgaaggagt cgagcgggtg 120
gcgcgcaaac tcgcccaccac cgaccgcgcg ttcaccgcg cgccctctcc 180
ccccgcacgc acgcgatcga cctcgacgag ccgctgcacg acaccgcgcg ggtcctcgac 240
gagctggagg acgtctacct ccgcgacgcc gtctacttcc accacccccc ctacctcgcc 300
cacctcaact ccccggtcgt cataccggcg ctgctcggtg aagcggctct gtcgcgcgtc 360
aactcctccc tggacacctg ggaccagtcg gccggcgcca cgctcatcga gcggaagctg 420
atcgactgga cctgcgcgcc catcgccctc ggcccggcgg ccgacggcgt gttcacctcc 480
ggcggcacc cgtccaacct caaggcgctg ctctcgccc gcaggaggcg gaaggccgag 540
gacttcgccc acctcgccgat ctctcgctcc gaggccagcc acttcagcgt cagggaagtc 600
gcgaacctgc tcggcctcgg ccccgacgcc gtctgtgcga tcccggtcga ccgcgcagaag 660
cggatgcaga ccgtcgccct cgcccgcgag ctggagcgct gcgcgcggga cggcctggtc 720
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tcgctcaagg cgcactcccg ttctggataa tgttttttgc gccgacatca taacggttct 11580
ggcaaatatt ctgaaatgag acgttgacaa ttaatcatcg gctcgataa tgtgtggaat 11640
tgtgagcgga taacaatttc acacaggaaa cagaatt 11677

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SEQ ID NO: 21      moltype = DNA length = 1443
FEATURE            Location/Qualifiers
source              1..1443
                    mol_type = genomic DNA
                    organism = Streptomyces pilosus

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SEQUENCE: 21
atgcgctcgc acctgctcaa cgacaccacc gcggagcact accgcgcctc cgtgaccgag 60
ggagtcgagc ggggtggcgc caaactcgcc accacagccc ggccgttcac cggcgctcag 120
gtcgacgccc tcgcgcccgc catcgacgcg atcgacctcg accggccctc gcacgacacc 180
accgcggtac cggaggaact ggaggacgtc tacctccggg acgcatccta cttccaccac 240
ccgcgctacc tcgcccacct caactgcccg gtcgtcatcc cggccctgct cggcgaggcg 300
gtgctctccg cgtcaactc ctcctgggac acctgggacc agtcggcggg cggcaccctc 360
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ggcgtgttca cctccggcgg caaccagtc aacctccagg cgtgctgct cgcgcgcgag 480
gagggcaaga ccggtgacct ggcgaaactg cgcgtcttcg cctccagagt cagccacttc 540
agcgtgaaga actcgcgaaa actgctcggc ctccggcccg acgcatgat ctgatccca 600
gtcgaccacg acaagcggct gcagaccgtc gccctcgccc cgcagctgga gcggtgcgtc 660
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gacgcgcgct accgctcggt tctgctcgcc tcaactgaagt accgcgacc catcgccg 840
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cactacctga agttcacct gctgaacccc gagaccacga cggacgacat ctccgcgctc 1380
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tga 1443

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SEQ ID NO: 22      moltype = DNA length = 1293
FEATURE            Location/Qualifiers
source              1..1293
                    mol_type = genomic DNA
                    organism = Streptomyces pilosus

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SEQUENCE: 22
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ctggagtcga agccggaact cgcctggcac gccgggatgt tctcgacgg cgcacacctc 180
cagacccctc tcgtgcgga cctggtcacc ctgcgcgacc ccacctccc ctactcctc 240
ctcaactacc tgaaggagaa gggccggctg tactcgttct acatacgca gaactctac 300

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ccgctgcgggg  tgcagtaacga  cgaactactgc  cgtctggggcg  ccgccaagct  gagcagcatc  360
cgcttcggga  cgacgggtcac  cgaggtgacc  tacgacgagg  ccgccgagct  gtacgcggtc  420
gcgacggcgcg  ccggtgagac  gtaccgcgcc  cggcacctgg  tgctcggcac  cggcacccgtg  480
ccgttctctc  cggagccgtg  ccgcggggtg  gacggcgggc  tcttcacaaa  cgcgcagtac  540
atgcagcgca  aggcgggaact  gctggagaag  aagtcgggtc  cggtcgtcgg  cagcgggtcag  600
agcgcgcgcg  agatctacta  cgaagctgctc  tccgagatcg  acgtccacgg  ctaccagctc  660
aactgggtca  cgcgttcgcc  gcggttcttc  ccgctggagt  acaccaagct  gaccctggag  720
atgacctccc  cggactacat  cgaactactc  cgcgcctcgc  ccgaggagac  ccgctaccgg  780
ctggagaagc  agcagaaggg  cctgttcaag  ggcatacaat  cggagctgat  cgacgccatc  840
ttcgacctgc  tctaccagaa  gaacgtcgag  agcggggacc  gtccgggtac  caccgcggctg  900
ctcaccaaat  ccagcctcac  cggcgcgcg  taccaggacg  gcgcgtacac  gctcggcttc  960
caccaggacg  agcagggcaa  ggacttcgag  atccgcaccg  agggcctggg  gctggccacc  1020
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ggccacggcg  gtttcgacgt  cgcgcccaac  tacgccatcg  acgtcacggg  ccggggggatc  1140
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ccgtaccgca  acgcgtccat  catccgtgag  ctgctgggca  ccgagtacta  cccggtcgag  1260
aagagcatcg  cgttcagga  gttcgccgta  tga  1293

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SEQ ID NO: 23      moltype = DNA  length = 540
FEATURE           Location/Qualifiers
source            1..540
                  mol_type = genomic DNA
                  organism = Streptomyces pilosus

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SEQUENCE: 23
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gacgtcgagc  gcgagtaact  gcgcacgcgc  gccacgagc  accaccacgc  ctaccctggg  180
ctgcacgacg  gcgaaccgcg  ctctctgatg  gagaagtacg  acccccggta  cgtcgaactc  240
gtcggcctgt  agacaccgca  gcccgcgagc  gtcggcatgc  acttctcggt  cgcgccacc  300
gaccgccccg  tccacggctt  cacccgggcg  gtcacacggc  ccgtgatgga  ggagctgttc  360
gcccaccccc  ccacccggcg  ggtggtggtc  gaaccggacg  tcggcaacaa  ggccgtgcac  420
gcccctcaag  aggcggctgg  ctctgtgcc  gagcgggaga  tcgacaagcc  ggagaagcgc  480
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SEQ ID NO: 24      moltype = DNA  length = 1773
FEATURE           Location/Qualifiers
source            1..1773
                  mol_type = genomic DNA
                  organism = Streptomyces pilosus

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SEQUENCE: 24
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ctgctgatcc  gcaaggcgct  cgcgcagttc  gccacgagc  ggctcatcac  gccgcaggac  120
tcggcgacg  gccggtacgc  cgtccacagc  gacgacggcc  ggacccggta  cagcttcaac  180
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cgcacggctg  cggaggtcac  ccgcgagtac  caggcgctcg  tgccggaact  cgcgcgaca  1620
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ttgcgcaaca  accggcagat  ggtcgacctg  tcggaccctc  ccggcgcgct  ccagctgggtc  1740
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SEQ ID NO: 25      moltype = AA  length = 480
FEATURE           Location/Qualifiers
source            1..480
                  mol_type = protein
                  organism = Streptomyces pilosus

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SEQUENCE: 25

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IERKLIDWTT	ARIGLGPAAD	GVFTSGGTQS	NLQALLLARE	EAKTGDLAKL	RVFASEVSHF	180
SVKSAKLLG	LGPDAVISIP	VDHDKRLQTV	ALARELERCV	EAGLVPMMAV	ATAGTTDFGS	240
IDPLPGIAEL	CGRYGAMMHV	DAAYGCGLLA	SLKYRDRIAG	IEHADSVTVD	YHKSFFQPV	300
SSAVLVRDTA	TLRHATYHAE	YLNPRRMVIE	RIPNQVDKSL	QTTRRFDAK	LWMTLRVMGA	360
DGIGGLFDEV	CDLAAEGWKL	LAADPRFDVV	VAPSLSTLVF	RYIPAAVTD	AEIDRANLYA	420
RKALPASGDA	VVAGTKVAGR	HYLKFTLLNP	ETTTDDISAV	LDLIAGHAEQ	YLGESLDRAC	480

SEQ ID NO: 26 moltype = AA length = 430
 FEATURE Location/Qualifiers
 source 1..430
 mol_type = protein
 organism = Streptomyces pilosus

SEQUENCE: 26

LALAEAPET	TYDFVGVLG	PFNLGLACT	EPISLNGVF	LESKPDAFW	AGMFLDGAHL	60
QTPFMSDLVT	LADPTSPYSF	LNLYLKEKGR	YSFYIRENFY	PLRVEYDDYC	RWAAAKLSSI	120
RFGTTVTEVT	YDEAAELYAV	ATAAGETYRA	RHLVLGTGT	PFLPEPCRG	DGGLFHNAQY	180
MQRKAELLEK	KSVTVVSGSQ	SAAEIYYELL	SEIDVHGYQL	NWVTRSPRFF	PLEYTKLTLE	240
MTSPDYIDYF	RALPEETRYR	LEKQQKGLFK	GINSELIDAI	FDLLYQKNVE	SGDRPVPTRL	300
LTNSSLTGAR	YQDGAYTLGF	HQDEQKDFE	IRTEGLVLAT	GYHYEPPAFL	DPVREIRIFD	360
GHGRFDVARN	YVIDVTGRGI	FLQNAHVH	SITSPDLGMG	PYRNASIIRE	LLGTEYYPVE	420
KSIAFQEFV						430

SEQ ID NO: 27 moltype = AA length = 179
 FEATURE Location/Qualifiers
 source 1..179
 mol_type = protein
 organism = Streptomyces pilosus

SEQUENCE: 27

MSIGTFTVRP	LDPLEDAELL	HRWVTDPKAA	FWMQDAALQ	DVEREYLRIA	AHEHHHAYLG	60
LHDGEPAPLM	EKYDPRVVEL	VGLYDPEPGD	VGMHFLVAPT	DRPVHGFTRA	VITAVMEELF	120
ADPATRRVVV	EPDVGNAKVH	ALNEAVGFVP	EREIDKPEKR	ALLSFCTREQ	FEARGASA	179

SEQ ID NO: 28 moltype = AA length = 590
 FEATURE Location/Qualifiers
 source 1..590
 mol_type = protein
 organism = Streptomyces pilosus

SEQUENCE: 28

MTLSDAVAHL	SPHRWARANR	LLIRKALAEF	AHERLITPQD	SGDGRYAVHS	DDGRTRYTFT	60
AARRALDHHW	IDPESITRHR	DDAERELAAL	DFFIELQKSL	GLSEILPVY	LEERISSTLSG	120
TCYKLTQKQV	RAGELIDAGF	QAVETGMTEG	HPCFVANNGR	LGFGVDEYRA	YAPEAAAPVK	180
LVLAAHRSR	AAFTAGYGIE	YESFVRQELG	EETVEGFHDV	LRAQDLDPGD	YLLIPVHPWQ	240
WVNKLSTVFA	AEIARRHLVC	LGEGDDEYLA	QQSIRTFENR	SHPGKHVYKT	ALSVLNMGMF	300
RGLSAAAMEA	TPAINDWLAQ	LVDSDPVLRA	TGLSIIRERA	AVGYRHLEYE	QATDRYSPYR	360
KMLAALWRES	PVPSLEDGES	LATMASLVHV	DHEGASVAAA	LIARSGLAPT	EWLRHYLRAY	420
YVPLHHSFYA	YDLAYMPHGE	NVILVLRDGV	VRRAIYKDIA	EEIVVMDADA	VLPPEVARIR	480
AEVPDDTKLL	SIFTDVDFCF	FRFLAANLAE	EGVLDEDFGW	RTVAEVTREY	QASVPELADK	540
FARYDMFAPE	FALSCLNRLQ	LRNNRQMVDL	SDPSGALQLV	GTLKNPLAEF		590

SEQ ID NO: 29 moltype = DNA length = 5102
 FEATURE Location/Qualifiers
 misc_feature 1..5102
 note = Vector Construct
 source 1..5102
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 29

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ctcaacgaca	ccaccgcgga	gcactaccgc	cgctccgtga	ccgagggagt	cgagcgggtg	120
gcggccaac	tcgccaccac	agcccgcccg	ttcaccggcg	tcacggtcga	cgccctcgcg	180
ccccgcacg	acgcgatcga	cctcgaccgg	ccccctgcacg	acaccaccgc	ggtactggac	240
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SEQ ID NO: 31      moltype = DNA length = 1293
FEATURE           Location/Qualifiers
source            1..1293
                  mol_type = genomic DNA
                  organism = Erwinia amylovora

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SEQ ID NO: 32 moltype = DNA length = 2355
 FEATURE Location/Qualifiers
 source 1..2355
 mol_type = genomic DNA
 organism = Erwinia amylovora

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SEQ ID NO: 33 moltype = AA length = 430
 FEATURE Location/Qualifiers
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 mol_type = protein
 organism = Erwinia amylovora

SEQUENCE: 33

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SEQ ID NO: 34 moltype = AA length = 784
 FEATURE Location/Qualifiers
 source 1..784
 mol_type = protein
 organism = Erwinia amylovora

SEQUENCE: 34

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SEQ ID NO: 35 moltype = DNA length = 3830
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 note = Vector Construct
 source 1..3830
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 35

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FEATURE           Location/Qualifiers
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                  mol_type = genomic DNA
                  organism = Pantoea agglomerans

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tcggtggcgt	gagtatgagg	ttggttctaa	aaacctgcgt	tcgtcctggg	cacgtggggc	7620
taagcgtgct	ttgggcatgt	attacataga	cgtgtatgta	cgtcgtgaaa	tggaaagaag	7680
actgtacaag	ctgcgcgggt	tggaaagcacc	ggaacgggtc	gaatcaaccc	gcgttgctgt	7740
tgctttggtg	aagcccgatg	attggaaaact	gattcagctc	gatttcgcgg	ttaggcagta	7800
cgttctcgat	tgctgtgata	aggctaagga	cgtggccgct	gcgcaacgtg	tcgctaatga	7860
gggtctggca	agtctgggtg	tggattccac	cccgtgcgat	atcggttatg	atgatgtgga	7920
cttggaacgc	gttctgccta	ctcatgggga	cgtactaaag	cgtgatctga	atgcggcggt	7980
gttcgcgggt	aatgagcaga	ctattctctc	caccactaaa	aagcggcata	aaacccggtc	8040
gatattttgt	gcgatgaatt	tatggtcaat	gtcgcggggg	caaaactatg	tgggtcttgt	8100
tggtggcgct	ccggaaaaac	attccgaagc	ccaaccttct	atagaaggcg	gcgggtggaat	8160
cgaatatctc	tgatggcagg	ttgggcgctc	cttggtcggt	catttcgaag	ggcaccataa	8220
actgccttaa	aaaaattacg	cccgcgcctg	ccactcatcg	cagtactggt	gtaattcaat	8280
aagcattctg	ccgacatgga	agccatcaca	gacggcatga	tgaacctgaa	tcgccagcgg	8340
catcagcacc	ttgtcgccct	gcgtataata	tttgcccatg	gtgaaaacgg	gggcgaagaa	8400
gttgtccata	ttggccacgt	ttaaatcaaa	actggtgaaa	ctcaccacag	gattggtctga	8460
gacgaaaaac	atatttctaa	taaacctttt	agggaaaatg	gccaggtttt	caccgtaaca	8520
gcgcacatct	tgcgaaatata	tgtgtagaaa	ctgcgggaaa	tcgtcgtggg	attcaactcca	8580
gagcgatgaa	aacgtttcag	tttgctcatg	gaaaacgggt	taacaagggt	gaacactatc	8640
ccatataccc	agctccacgt	ctttcattgc	catacggaac	tccggatgag	catttcacag	8700
gcgggcaaga	atgtgaataa	aggccggata	aaactttgtc	ttatttttct	ttacggctct	8760
taaaaaggcc	gtaatatcca	gctgaacggg	ctgggttatag	gtacattgag	caactgactg	8820
aaatgcctca	aaatgttctt	tacgatgccca	ttgggatata	tcaacgggtg	tatatccagt	8880
gatttttttc	tccattttag	cttccttagc	tcctgaaaaa	ctcgtcgaag	ctcggcggtg	8940
ttgtcctact	caagctgac	cgacaaaatc	cacacattat	cccaggtgtc	cggatcgggtc	9000
aaatagcctg	ccagctcata	gaccgtatcc	aaagcatccg	gggctgatcc	ccggcgccag	9060

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ggtaggttttt cttttcacca gtgagacggg caacagctga ttgcccctca ccgectggcc 9120
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atgggtggtta acggcgggat ataacatgag ctgtcttcgg tatcgtcgtg tcccactacc 9240
gagatatccg caccaacgcg cagcccgagc tcggtaatgg cgcgcattgc gcccagcgcc 9300
atctgatcgt tggcaaccag catcgacgtg ggaacgatgc cctcattcag catttgcatg 9360
gtttgttgtaa aaccggacat ggcactccag tcgccttccc gttccgctat cggtgaatt 9420
tgattgcgag tgagatatgt atgccagcca gccagacgca gacgcgccga gacagaactt 9480
aatggggccc ctaacagcgc gattttgctg tgacccaatg cgaccagatg ctccacgccc 9540
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tccagcggat agttaatgat cagcccactg acgctgttgc cgagaagatt gtgcaccgcc 9720
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tggtggcct gggtccaccg gcgggaaacg gtctgataag agacaccggc atactctgcg 10020
acatcgata acgttactg tttcacattc accaccctga attgactctc ttcggggcgc 10080
tatcatgcca taccgcgaaa ggttttgac cattcgatgg tgtcaacgta aatgcgcgtt 10140
cgcttcgcg cgcaattgc aagctgatcc gggcttatcg actgcacggt gcaccaatgc 10200
ttctggcgct cggagcccat cggaaagctgt ggtatggctg tgcaggtcgt aaatcactgc 10260
ataattcgtg tcgctcaagg cgcactcccg ttctggataa tgttttttgc gccgacatca 10320
taacggttct ggcaaatatt tgaatatgag ctgttgacaa ttaatcatcg gctcgtataa 10380
tgttggaat ggtgagcgga taacaatttc acacaggaaa cagaatt 10427

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SEQ ID NO: 43      moltype = DNA length = 1344
FEATURE            Location/Qualifiers
source              1..1344
                    mol_type = genomic DNA
                    organism = Corynebacterium xerosis

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SEQUENCE: 43
gtgacgacg aggatccca cgacctcatc ggcacgagg tcggcccctt cgggctggga 60
ctggctgcgc ttgcgacacc gctgcgggac gtgcagcgcc tgttccctga ccaggccgac 120
aggttctcct ggcacccggg actgatgatc gagggcacca ccctccaggt ccccttctc 180
goggacctcg tgacgatggc ggatccacc tcgcctact ccttctcgc cttctcaag 240
cagaccggca ggtctatccc cttctacatc cgggaatcct tctaccgct gcgcagcgag 300
tacgacgagt actgcccgtg ggtcgccgca cagctggact gcctgcgctg gcggcgctcg 360
gtggtcgccg tggaggagga gggcgatcac ctgctcgtca cggccgcccgt gctcgatgca 420
accggggctg tgatcgctg cgagcacctg cgcggccgcc acgtggctct cgcgcgtcgc 480
accgcgcgcg accttccgcg gccctcgccc gagctgatcg gcccgccggg tgggtgctca 540
cgacacgagg cccagccccc cgtgctgcac agcgcgcgact acctccggcg gaaggacgag 600
ctgctcgccg ccggatccgt gacgggtggt ggcagcgccc agtccgcgc agagatctat 660
cgggacctgc tccaggaggg cggggccgac gggctgcggg tggactgggt gaccggttcc 720
ccgcggttct tcccatgga gtacacgaag ctacacctcg agatgacctc gccggagtag 780
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gcccgcaggg actccgggag cggacggatc gtctgggacc tcgagcatcc gctcacgggc 1020
tcggtcgccg agcatcgcac cggcgccgtg gtgcgcgaga cgggctaccg gccccgctcc 1080
accgagctgc tcgacacggc cggggacggg atccgcacag acgacgcggg ccgggctgac 1140
gtcaccccgc actacaccgt cgaccgcgag gagcggtacc atgtgctggg cggcgaggag 1200
cacacccacg gcgtcaccgc cctgacctc ggcttcggca ccttgccgc ctcgctgatc 1260
ctcgccgcgc tcaccggggc ggaagcgtat ccgatcgagc gtcgcacgc cttccagacc 1320
ttcgccctcc ccgcgcgccg ctga 1344

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SEQ ID NO: 44      moltype = DNA length = 2430
FEATURE            Location/Qualifiers
source              1..2430
                    mol_type = genomic DNA
                    organism = Corynebacterium xerosis

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SEQUENCE: 44
atgaacctga tgaacccgaa cgaccggtt gacctgagcg tcgaaccgct ccgtcctgcg 60
gaccacaccg gcctgctgca cagctgggtg agcacccgcg gcgcccgttt ctggggcatg 120
gtcgaccatt cgcccgacac ggtgcgcgcc tacctcgacg ccgtcgccgc ctccagcac 180
gagaacggcc ggatcgccct gcgggacggg gtcccgtcgc tctacgtcga gacctatgat 240
cccgcccgac tgctgcgcgc cgggggtgct catccggaac ccggcgatct cggcatgca 300
ctcctcatcg cgccgcggga cggcccgacc gagcacggcc tcaccgcccg ggtgctggca 360
ggcgtcgtgc gctggtgctt cgagaaccgt ggcgcgcccc ggatcgtcgt cgaacccgac 420
gaacgcaaca gcgcagtgct cgccaagaac gccgcagccg gtttcgcgt gctgcgcacc 480
gtcgaaactg ccgagggatc gggcacgaag cgcgcggtgc tgagcatctg caccggggcg 540
gacttcgcgg cttcccgcgt cggggaccgg cggcagcgca aggaactccg gggcgccgcc 600
ggagccgacg acgcgactgc ggaagcgcca catctcgccg ctccgctcgc cccgacgtcc 660
catctgcgcg ccgacgtagc cgagcggggc caccggcacc tgatcgccaa ggccctcgcc 720
gagctcagcc acgaacgagt cctcgccccc gagcgctcg gcgacgatcg cttcgccctg 780
gcagtcctcg gcaccgacgt acgctacgag ttccggggcc gcgtgctccc gctcagagac 840
tggtggtggg acccgccctc gatccggcgc ctcgaggacg gtgcccgcac ggcgctggac 900
gtgcagatcc tcgtgttgga gctgcaggat ccgctcgcca tcgcccagca tctgctctcc 960

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acctacctcg aagagctcgc ctccaccttc gccgcccgt gccgcagcgt ggacgaggcc 1020
cgctcgccggc agcggccgctc ggcagagggc ctgctcacgg cggacctccc ggacgtcgag 1080
gccgccatga cggagggcca cccgggcttc ctgcggaaca acggggcgcg cgggttctcc 1140
ctgcccgaact accgcgccta cgcaccggag cagggtcggc acacgcgcct gcagtgggtc 1200
gcggtgcacg ccgcgcgcag caccctgtcc ctgcgagagg ggtgggacga ggcgcgcgat 1260
gcccgttgga gcctctcgtc cgaggagcgc gactccttcg cggaccgtct ggcgcgcgag 1320
ggcctggacc cgtcggacta ccacctgtcg cccctgcctc cctggcaggc agagcatcgt 1380
ctgcccatac ccttcgcccc ggacatcgcc cgtggggacc tcgtgcccct cggggcgagg 1440
ggcgacgaga tgcagcccca gcagtcctcg cgcaccttcc tcaaccgcac gcgagcctgg 1500
gcgcctatg tgaaggctgc cctcgcgac cagaacatgg ggttcctgcg cggcctctcg 1560
ccgcagtaga tgcgggacac ccccacgatc aacgactggc tcgcgcgcgt cgtggacagc 1620
gaccccgagt tcgcgggcgg cggcttcgcc gtccctgcggg agcgggcagc actgggctac 1680
accggggagc cctaccaccg cagcccgccc acgaatcccc accgcaagat gctcgcggcg 1740
ctgtggcggt agagccccct gccgtgctc gagccggggg agcgggcctt cagcatggcg 1800
gcgtcctctc acccgcatca cgcgcgcacg gcgtcctgta gcacgatgat ccgcacctcg 1860
gagctgcggc cggaggtctg ggtcgccca gacctgcgcg cctacctccg cccgctcgtc 1920
catgcgctgc tcgcgcacga cgtggtggtc atgccgcacg gcgagaacct catcctgcgg 1980
gttcaggagc agcaggtggt cggggcattc ctcaaggaca tcggggaggga gatggcggtg 2040
ctggggccac gcccgctgac tgcggtgatg gagcgggtac gggccgtggt ccccgcgag 2100
gacaaggccc tgtcgatctt caccgacgtc ttcgacgggg tgcctgcggca tctcgcgggg 2160
atcctcgacg ccgacggggg gcttcgccgc gatcgcttct ggcgggtcgt ggcgggcac 2220
ctggacgagt acgagggcga ccatccggat ctgcgcgggg ggatctccgg ggacatcgac 2280
ctgcgtgcgg gcacctctgc gcaactcgtg ctgaaccgcc tcgagctgcg gaacacctg 2340
cagatggtcg acctcggcga ccaggcctcg tcgtcgtctc tcgcggggga catggacaac 2400
ccggtggctc gggcgctccg cctgcctga 2430

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SEQ ID NO: 45      moltype = AA length = 447
FEATURE
source            Location/Qualifiers
                  1..447
                  mol_type = protein
                  organism = Corynebacterium xerosis

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SEQUENCE: 45
VHDEEIHDLI GIGIGPFLGL LAALADPLPD VDAVFLDQAD RFSWHPGLMI EGTTLQVPFL 60
ADLVTMADPT SPYSFLAFLK QTGRLYPFYI RESFYPLRSE YDEYCRWVAA QLDCLRWRRR 120
VVAVEEEGDH LLVTAAVLDA TGAVIGCEHL RGRHVALAVG TAPHLPPPLA ELIGPAGGAP 180
ADEAPAPVLH SADYLRRKDE LLAAGSVTVV GSGQSAAEIY RDLLQEGGAD GLRVDVWTRS 240
PRFFPMETK LTLMTSFEY TDLHRSLEPG TRDRLAREQR ALHKGISADL IDEIHELLYR 300
LSAGGREVPS SLLTGAAVRA ARDSDSGSRI VLDLEHPLTG SVAEHRTGAV VAATGYRPRS 360
TELLAPLGDR IRHDDAGRLD VTRDVTVDRE ERIHVLGGEE HTHGVTAPDL GFGLTLRASVI 420
LAALTGREPY PIERRIAFQT FGLPAAR 447

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SEQ ID NO: 46      moltype = AA length = 809
FEATURE
source            Location/Qualifiers
                  1..809
                  mol_type = protein
                  organism = Corynebacterium xerosis

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SEQUENCE: 46
MNLMPNDPL DLSVEPLRPA DHTGLLHSLW STPRARFWGM VDHSPAQVRA YLDAVAASEH 60
ENRIGLRDGT VPLVYVETD PARLLPAGVL HPEPGDLGMH LLIAPPDGP EHLTAAVLA 120
GVVRWCFENR GAARIVVEPD ERNSAVLAKN AAAGFRVLRT VELPEGSGTK RAVLSICTRA 180
DFAASPLGDR RQRKELRGGA GADDATAERA HLGAPVAPTS HLRADVAERA HRHLIAKALA 240
ELSHERVLLP EALGDDRFAL AVPGTDVRYE FRARVLPLEH WLVPASIRR LEDGAPTALD 300
VQILVLELQD PLGIADLLS TYLEELASTF AARCATLDEA RRGERPSAEG LLTADLPDVE 360
AAMTEGHPGF LANNGRIGFS LPDYRAYAPE QGRHTRLQWV AVHAARSTLS LGEGLDAAH 420
ARWSLSSEER DSFADRLAGR GLDPSDYHLL PLHPWQAEHR LPITFAPDIA RGLVPLGAG 480
GDEMOPQQSL RTFLNRTRAW APYVKVALAI QNMGFLRGLS PQYMRDTPTI NDWLALVDS 540
DPEFAGGGFR VLRERAALGY TGDAYHRTPA TNPHRKMLAA LWRESPLPL EPGERAFTMA 600
ALLHRDHDGT ALVSTMTIRS ELPAEVWVAQ YLRAYLRPLV HALLAHDVVV MPHGENLILR 660
VQEQQVVGAF LKDIGEEMAV LGHRPLPAVM ERVRAVVPGE DKALSIFTDV FDGVLRHLAG 720
ILDADGVLPD DRFWRVVAGI LDEYEADHPD LARGISGDID LRAGTFAHSC LNRLQLRNTL 780
QMVLDLGDQAS SLLFAGDMDN PVARASVPA 809

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SEQ ID NO: 47      moltype = DNA length = 3922
FEATURE
misc_feature      Location/Qualifiers
                  1..3922
                  note = Vector Construct
source            1..3922
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 47
aagcttgcca cgggggtgga ccgatgagcg cccgggccac gccggaggcg gagagcccgg 60
aaccgcgggt gcacgacgag gagatccacg acctcatcgg catcgggatc ggcctctcgg 120
ggctgggact ggctgcgctt gccgacccgc tgcgggacgt cgacgcctgt ttctggacc 180
aggccgacag gttctcctgg caccggggac tgatgatcga gggcaccacc ctccaggctc 240
ccttcctcgc ggacctcgtg atccacaccc gccctactcc ttctcgcct 300
tcctcaagca gaccggcagg ctctaccctc tctacatccg ggaatccttc taccgcgtgc 360

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gcagcgagta cgacgagtag tgccgctggg tcgcgcgaca gctggaactgc ctgcgctggc 420
ggcgctcggtt ggtcgccgtg gagggaggag gcgatcacct gctcgtcacg gccgcgctgc 480
tcgatgcaac cggggctgtg atcggtctgc agcacctcgc cggccgcac gtggtctctc 540
ccgtcgccac agcgccgcac ctcccgccgc ccctcgccga gctgacggc cggccggtg 600
gtgctccagc agacgagggc ccagcccccg tgctgcacag cgcgcactac ctccggcgga 660
aggacgagct gctcgccgcc ggatccgtga cgggtggtggg cagcgggcag tccgcgcag 720
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ccggttcccc gcggttcttc cccatggagt acacgaagct caccctcgag atgacctcgc 840
cggagtacac cgacctgcac cgtcccttc ccgaggggac gcgcgacgc ctggcgcggg 900
agcagcgtgc cctgcacaag ggcacacgc cggacctgat cgacgagatc cacgagctcc 960
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tcacgggctc ggtcgccggg catcgccacg gcgcgctggg cgcgcgcacc ggctaccggc 1140
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ggctcgacgt caccgcgcac tacacctcgc accgcgagga gcggatccat gtgctgggcg 1260
gcgagggaca caccacggc gtcaccgccc ctgacctcgg ctccggcacc ttgcgcgcct 1320
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tcacgacctt cggcctcccc gccgcgcgct gaccaggagc gtccccatga acctgatgaa 1440
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cgtagccgag cgggcccacc ggcaacctgat ccgcaaggcc ctgcgcgagc tcagccacga 2160
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cgacgtacgc tacgagttcc gggcccgctg gctcccgctc gagcaactggc tgggtgagcc 2280
ggcctcgatc cggcgctcgc aggaacgggtg ccgcacggcg ctggacgtgc agatccctct 2340
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cgctacgcca ccggagcagg gtcggcacac gcgcctgag tgggtcgcg tgcaacgcgc 2640
gcgcagcacc ctgtccctcg gcgaggggct ggcagaggcc gcgcgctccc gctggagcct 2700
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ggtctgggtc gcccagtagc tgccgcgcta cctccgccc ctcgtccatg cgtgctcgc 3360
gcacgacgtg gtggtcatgc gcacggcgga gaacctcctc ctgcggttc aggagcagca 3420
ggtggtcggg gcatctctca aggacatcgg ggaggagatg gcggtgctgg gccaccgccc 3480
gctgctcgcg gtgatggagc gggtagggcg cgtggtcccc ggcgaggaca aggccctgtc 3540
gatcttcacc gacgtcttcg acggggtgct gcggcatctc gcggggatcc tcgacgcgca 3600
cggggtgctt cccgcgcgac ccttcggcg ggtcgtggcg ggcacccctg acgagtacga 3660
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cttcgcgcac tcgtgcctga accgcctgca gctcgggaac accctgcaga tggctgacct 3780
cggcgaccag gccctgctcg tgctcttcgc cggggacatg gacaaccccg tggctcgggc 3840
gtccgtccct gccctgaggtc cggggcgggc gcgctcagct ctccaggagc cgcgcgcgcg 3900
ccgcgagggg gtcgccaagc tt 3920

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SEQ ID NO: 48      moltype = DNA length = 10507
FEATURE            Location/Qualifiers
misc_feature       1..10507
                    note = Vector Sequence
source             1..10507
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 48
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accgcgggtg cagcagagg agatccacga cctcatcgcc atcgggatcg gcccttcgg 120
gtggggactg gctgcgcttg accaccgct gccggacgtc gacgcccgtg tctggacca 180
ggccgacagg ttctcctggc acccgggact gatgatcgag ggcaccaccc tccaggtccc 240
cttctcgcg gacctcgtga cgatggcgga tcccacctcg ccctactctt tctcgcctt 300
cctcaagcag accggcagcg tctaccctt ctacatccgg gaatcctct acccgctgcy 360
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cctggagaaa tgctgaccaa agagagccgc acagtactcg attttctact gatgctttgt 2040
tccgtcgggc aacattaccc cggttttgaa acggatattc acggcgcgaa acaggacgaa 2100
gacggcgctt accgcgtacg agtcctaaaa atggcgggat aa 2142

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SEQ ID NO: 50      moltype = AA length = 713
FEATURE           Location/Qualifiers
source            1..713
                  mol_type = protein
                  organism = Escherichia coli

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SEQUENCE: 50
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EYSLDLCSDI NQLNEYLPFY AFINTHSTMD VSVQDMRMAL WFFEYALGQA EDIAIRMROY 120
TDEYLDNITP PFTKALFTYV KERKYTFCTP GHMGGTAYQK SPVGCLFYDF FGNITLKADV 180
SISVTELGSL LDHTGPHLEA EBYIARTFGA EQSYIVTNGT STSNKIVGMY AAPSGSTLLI 240
DRNCHKSLAH LLMMNDVVPV WLKPTRNALG ILGGIPRREF TRDSIEEKVA ATTQAQWPVH 300
AVITNSTYDG LLYNTDWKQ TLDVPSIHFD SAWVPYTHFH PIYQKSGMS GERVAGKVIF 360
ETQSTHKMLA ALSQASLIHI KGEYDEEAFN EAFMMHTTTS PSYPIVASVE TAAAMLGRNP 420
GKRLINRSVE RLFHRKEVQ RLREESDGWF FDIWQPPQVD EAECWPVAPG EQWHGFNDAD 480
ADHMFDPVK VTILTPGMD E QGNMSEEGIP AALVAKFLDE RGIVVEKTGP YNLLFLFSIG 540

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IDKTKAMGLL	RLTEFKRSY	DLNLRKINML	PDLYAEDPDF	YRNMRIQDLA	QGIHKLRKH	600
DLPGLMLRAF	DTLPEMINTP	HQAWQRQIKG	EVETIALEQL	VGRVSANMIL	PYPGVPPLM	660
PGEMLTKESE	TVLDLFLMLC	SVGQHYPGFE	TDIHGAKQDE	DGVYRVRVLK	MAG	713

SEQ ID NO: 51 moltype = DNA length = 2433
 FEATURE Location/Qualifiers
 misc_feature 1..2433
 note = Vector Construct
 source 1..2433
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 51

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tagctggcg	ttaccctgcg	aatgtccaca	gggtagctgg	tagtttgaaa	atcaacgccc	180
ttgcccttag	gattcagtaa	ctggcaagga	aggattttcc	aggaggaaca	catgaacatc	240
atcgctatca	tgggcccctca	cgggtgttttc	tacaaggatg	agccaatcaa	ggagctggaa	300
tcgcgactag	ttgcacaggg	ctttcagatc	atctggcccc	agaactccgt	tgaccttctc	360
aaattcatcg	agcacaatcc	tcgcatttgt	ggtgtgattt	tgactggga	cgagtactct	420
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aagtacactt	tctgcacccc	aggccacatg	ggcggcaccg	cctaccagaa	gtccccagtc	720
ggatgcctct	tctacgactt	cttcggcggt	aacactctta	aggcagatgt	ctccatttcc	780
gtcacccagt	tgggctctct	gctggaccac	accggccctc	acctggaggc	agaagagtag	840
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aacaagatcg	ttggtatgta	cgcagctcct	tctggctcca	ccctgttgat	cgaccgcaac	960
tgtaacaagt	ccctgcgcga	tcttcttatg	atgaacgatg	tggtccctgt	atggctgaag	1020
ccaacccgta	acgctctggg	cactcctggc	ggtagccccc	gtcgcgaggt	caccctgtat	1080
tcattcgagg	aaaagggtgc	agccactacc	caggcacagt	ggcctgtcca	cgctgtcatt	1140
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gttccttcca	ttcacttcga	cagcgcatgg	gttccttaca	ctcacttcca	ccaatctatc	1260
cagggtgaag	ccggaatgtc	cgccgagcgt	gtcgctggca	aggttatctt	cgaaacccaa	1320
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atgcgcaccc	agacctcgc	acagggcctc	cacaagctca	ttcgcaagca	cgacctgcca	2040
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atgctcacca	aggagtcccg	caccgtattg	gacttccctc	tcattgtgtg	ctctgttggc	2280
cagcactacc	caggcttcga	gaccgacatc	cacggcgcta	agcaagatga	agacggcggt	2340
taccggttc	cgctccttaa	gatggcaggc	taacttgcca	gagcggcttc	cgggcgagta	2400
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SEQ ID NO: 52 moltype = DNA length = 1046
 FEATURE Location/Qualifiers
 misc_feature 1..1046
 note = Vector Construct
 source 1..1046
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 52

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ccttcaagta	caccccgaa	cgcggtcgcc	tcgtcagcgc	catcgaggtt	gaggaagatg	180
acgagatctt	cgcatcacc	tccgcggcgc	gcgttggtcg	caccgaagtc	aagcagatcc	240
gaccatcctc	ccgtgcaaca	atgggtgttc	gactgggtcaa	cttggaaaga	gggtgtagaac	300
tgcttgccat	cgacaagaa	gtcgaagacc	aggcggaagc	atccgcagaa	gcagtacgaa	360
agggtgcagt	cgaaggacca	gcatccaaga	ctgctgccga	agaaaccgac	tccgttgaca	420
acggatccga	cgaaaacggc	gaggaataat	ttagggcatc	ccgagaagta	tccatcacc	480
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tgagcatttc	cgcacttctg	ggagctatcg	gagcgattac	cgctcgctgtg	cttgaccat	720
tggttgcaat	catctacaac	tcgattgttg	acctcttcgg	tggaactcag	attcaactgc	780
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gaacagggga tttgttgat tgttaacttg aaggtagagt cttatctcgt tcttagaggg 900
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gcaagcagga tgtcaggagg cgcgcc 1046

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SEQ ID NO: 53      moltype = DNA length = 1028
FEATURE
misc_feature      1..1028
                  note = Vector Construct
source            1..1028
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 53
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tcattggaag gctgtgtgta ctttcgacaa caagatttgg gcaaaatggt ggtaaaatc 180
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gggtgcctgc aagtgcgata ctggtgacaa gtgtcgtgga tccgaaagcc agaataaaag 300
gcttgacccc aacatggatc aggttctgga tttttacgcc acaccgaga ccaaacattg 360
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caacagtggg gcgcaaaact accatcgcca agaatacaag gataaatagg ggaacgaccg 480
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cagcaatggg tgcaagtagg agtactcggg cgagtctcac cacaactgca acacctaattg 600
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aaggaaatcat cagcgttccg aaaataacaa caagtgcac cgcagtaacg acctcttctt 780
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tccccaaaat accaccggca acgatgcaca caatcaccgc cagcatgggg aaaccaatgg 960
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ccggggat 1028

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SEQ ID NO: 54      moltype = DNA length = 7727
FEATURE
misc_feature      1..7727
                  note = Vector Construct
source            1..7727
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 54
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1. A method for processing a layer element comprising at least one base body with a lower side and an upper side opposite the lower side,
a cover layer that is at least partially transparent to visible light, and at least one adhesive layer arranged between the cover layer and the upper side, t
the method comprising
heating the adhesive layer to an adhesive layer temperature; and
cooling the lower side of the at least one base body to a lower side temperature.

2. The method according to claim 1, wherein the layer element is a solar module.

3. The method according to claim 1, wherein the adhesive layer temperature is at least 180° C.

4. The method according to claim 1 wherein the lower side temperature is at most 150° C.

5. The method according to claim 1 wherein the adhesive layer is heated by infrared radiation and/or magnetic induction and/or microwave radiation.

6. A device for carrying out a method according to claim 1, comprising:

at least one heating device for heating the adhesive layer of the layer element to the adhesive layer temperature; and

a cooling device for cooling the lower side of the base body of the layer element to the a lower side temperature.

7. The device according to claim 6, further comprising a workbench for supporting the layer element, wherein the cooling device is arranged in the workbench.

8. The device according to claim 6 wherein the cooling device comprises at least one fluid channel through which a cooling medium is directable.

9. The device according to claim 6 wherein the cooling device has at least one fan through which air is directable onto the lower side of the layer element.

10. The device according to claim 6 further comprising a first temperature sensor for detecting the adhesive layer temperature and/or a second temperature sensor for detecting the lower side temperature.

11. The device according to claim 10, further comprising an electric control unit configured to control the heating device and/or the cooling device depending on a detected adhesive layer temperature and/or a detected lower side temperature.

12. The method of claim 2 wherein the solar module metallic conductor tracks which are arranged on the upper side of the base body and are covered by the cover layer.

13. The method of claim 3 wherein the adhesive layer temperature ranges from 200° C. to 400° C.

14. The device according to claim 11 wherein the electric control unit is an electronic data processing device

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